HEAR

request audio





+ history turns

Gipformer RNN-T 68M

base	15.2%	
v2	8.7%	
v3	**7.7%**	

WER on val audio · sherpa-onnx, CPU

*"**trả tiền điện** bằng **ví điện tử** giúp mình ví mà **không được** thì trừ vào **tài khoản ngân hàng** nhé"*

misheard **không được** · said **không đủ**

ASR fine-tuning data (TRAIN audio only)

- **number labels** spelled from their value: 368 / 368 kept
- **augmentation**: speed 0.9–1.1, noise, SpecAugment
- **TTS**: 1,202 cloned-voice sentences, new names and numbers
- **final**: + dev and validation audio (all in)

DECIDE

Prompt E1P (≈ 6k tokens, ≤ 16k)

- **context**: time, user profile, installed apps
- **weekday table**: "thứ bảy tuần sau" → a date
- **Vietnamese app names**: "bên ví" → ewallet
- **recovery**: empty search → shorter query
- **72%** of the tokens are tool schemas

- rules + recovery hints
- app names (vi) + schemas
- context + weekday table
- history
- request (transcript)

Qwen3.5-4B

LoRA p1	`pay(bank)`
LoRA e1	`pay(bank)`
LoRA synth	`pay(ewallet)`

vote

per turn

2 / 3 → `pay(bank)`

tie → p1

val 70.9 → 72.7

tool result → next turn

ACT

Scored on

- **final state** = gold state, then
- **call F1** over the ordered calls
- **≤ 30 s** per sample, offline
- **LLM < 5B**, total < 6B params

3 calls, in order

`+ state_hash`

missing slot → **ask back**

ToolEnv *one per sample, every call counts*

1. `bill_payment.list_bills()` điện **845.000đ**, bill_02
2. `ewallet.get_balance()` **713.606đ** < 845.000đ
3. `bill_payment.pay(bill_02, source="bank")`

stop when the vote says: no call

- schema
- x-ref IDs seen
- no repeats
- retry cap: 2 empty